Power Flow Analysis

A Linear Solution to Economic Electric Grid Scheduling

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Final Project for CO 370 - Deterministic OR Models
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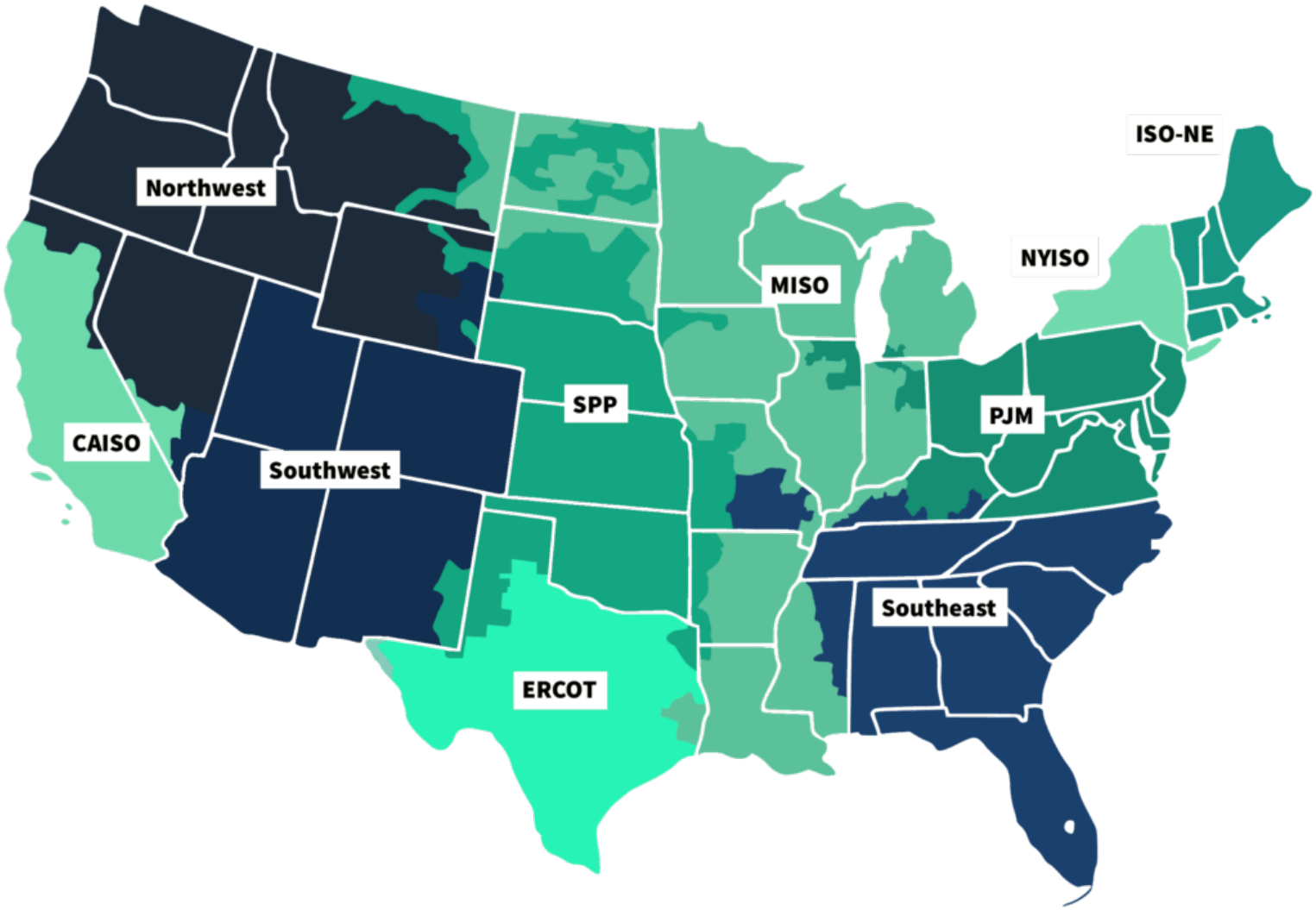
The US Power Market



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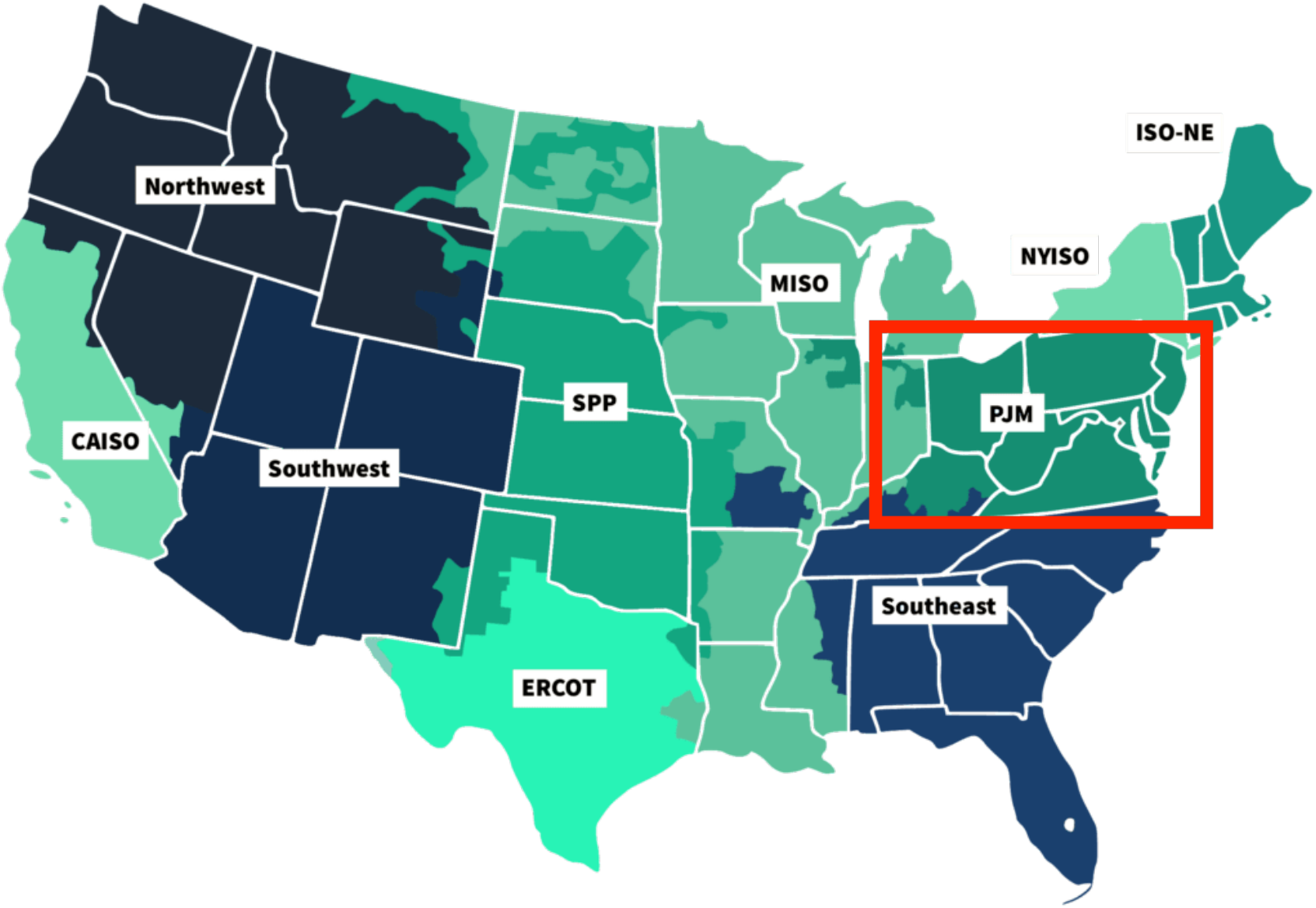

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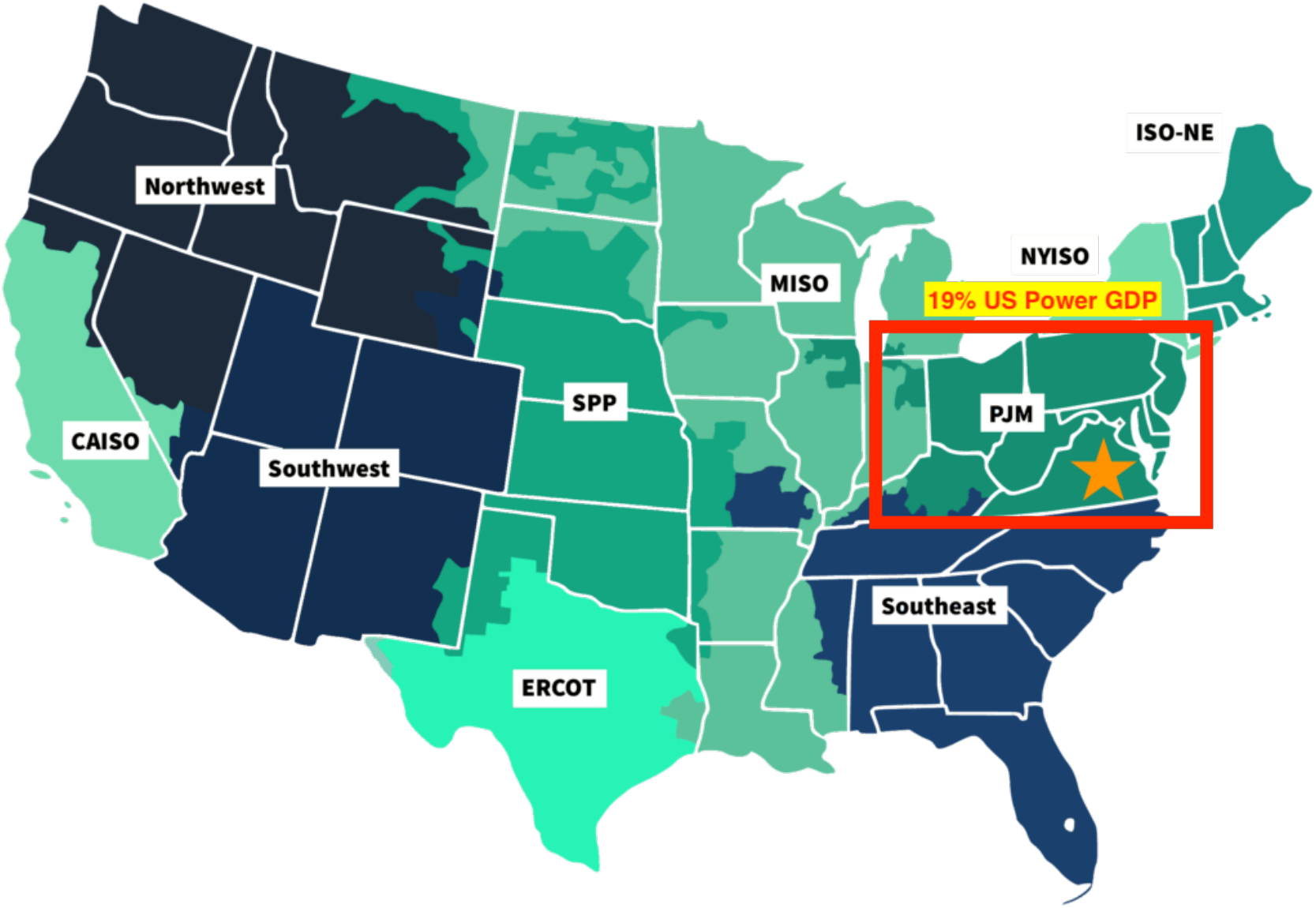

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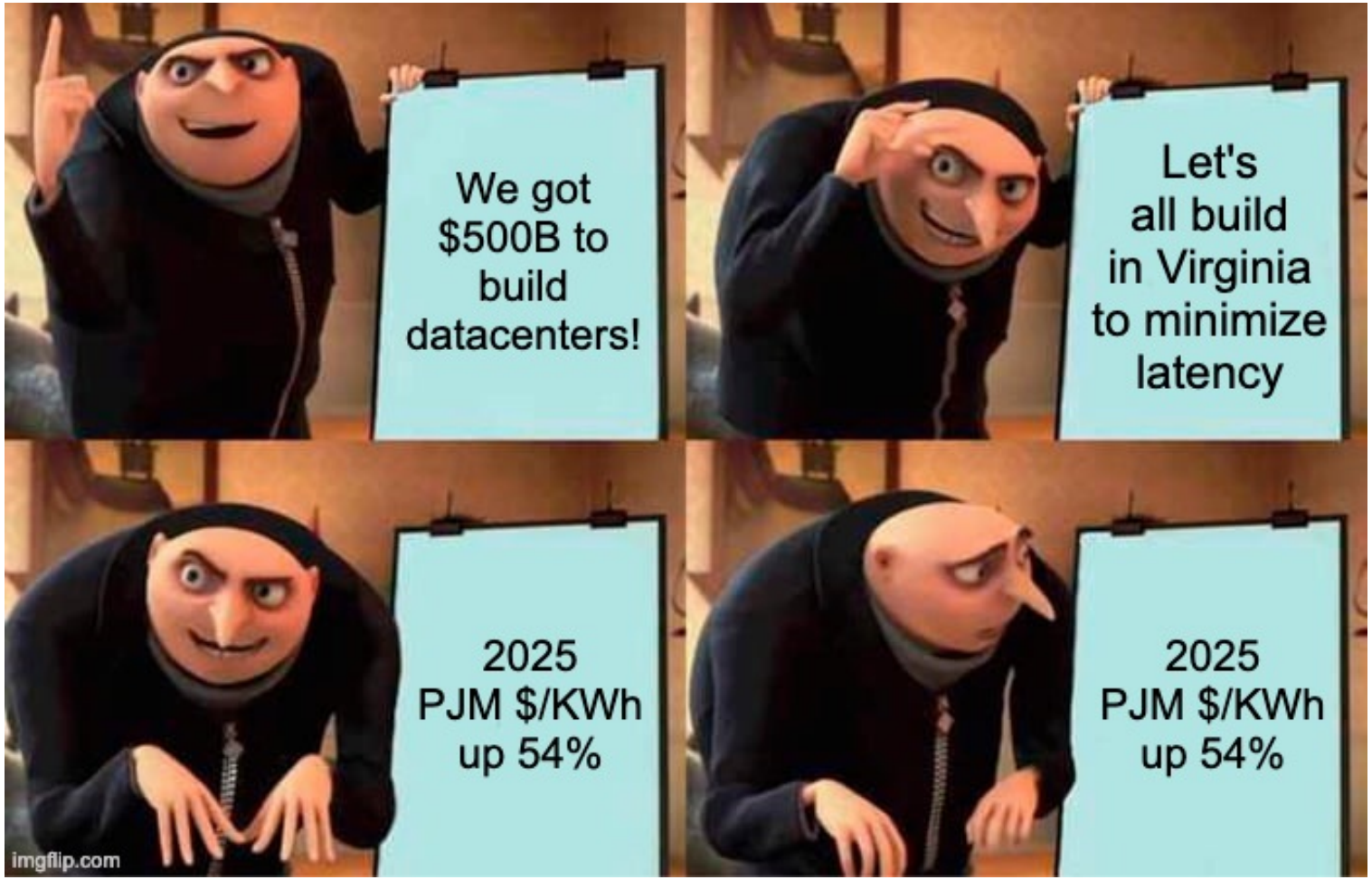
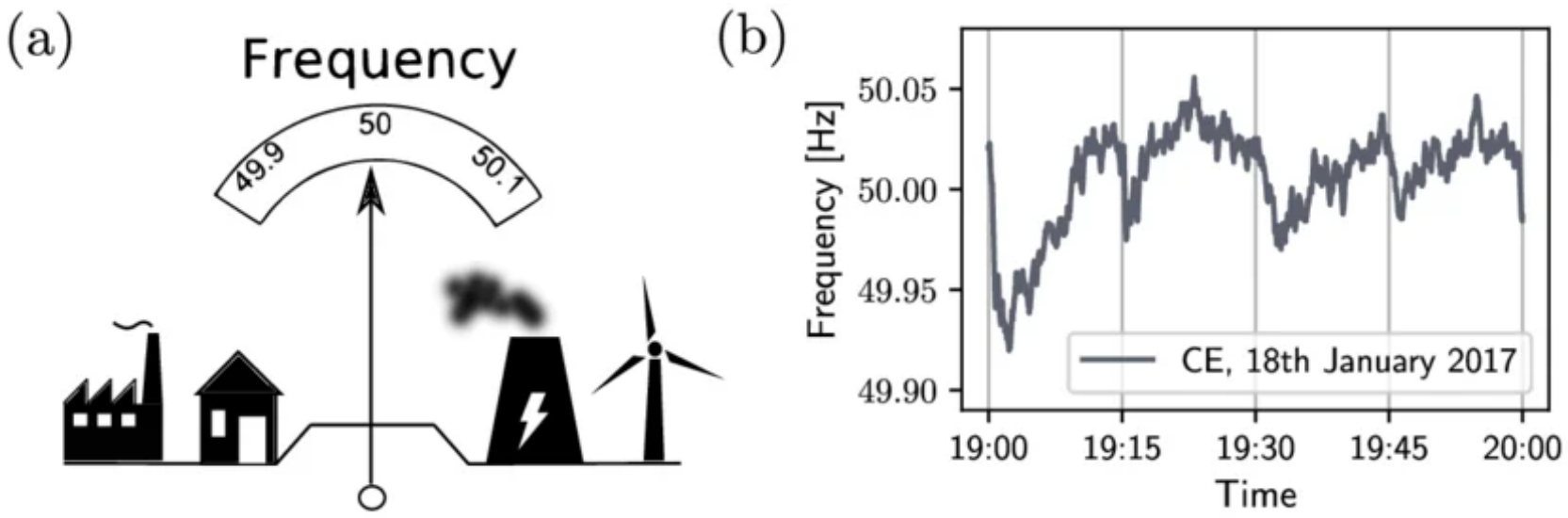


Source: Utility Dive on PJM capacity markets

Presentation Overview

1. Problem Description
2. Small Example
3. DA Clearing LP Formulation
4. FTR Clearing LP Formulation
5. IEE 118-Bus Large-Scale Example
6. Results Analysis

The North American Power Grid



- The power grid is a network of generators, loads, and power lines that are constantly trading with each other to supply electricity.
- *Unlike other markets*, power is delivered instantly and is subject to transmission and power-balance constraints due to physical limits.
- Regional Transmission Operators (RTOs) ensure economic dispatch of power by telling participants how much power to use/generate.

Aside: Electricity 101

- Electricity doesn't care about what you want.
- 1 MW injected into a single point in the grid will spread out across the network and intermingle with power from other generators.
- The amount of power that flows through each transmission line is defined by which gens and loads are authorized to inject/withdraw.

Takeaway 1

Electricity can't easily be "routed" through transmission lines. It flows relative to physical conditions of the grid. The percentage of each MW that goes through each transmission line is its shift factor.

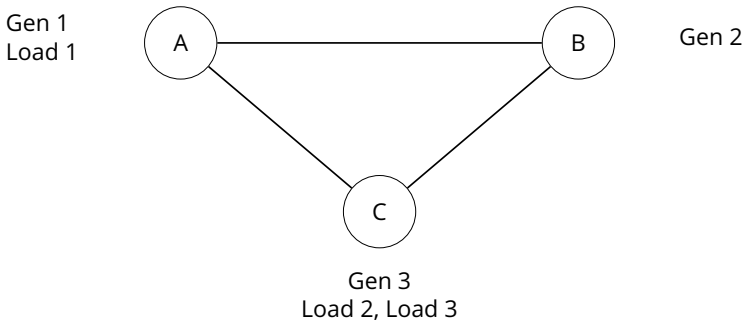
The Market-Clearing Problem

- In the Day-Ahead (DA) market, generators submit offers and loads submit bids for the next 24 hours.
- The RTO clears the auction to obtain a feasible generation and consumption schedule maximizing revenue (to maximize grid utilization).
- Additionally, the RTO places a (N-1) contingency constraint, requiring any cleared solution to also be feasible if any one line goes down.
- We study a DC approximation of optimal power flow, then extend the model to clear Financial Transmission Rights (FTRs).

Takeaway 2

There are many feasible (and even more) infeasible solutions to the DA clearing problem. We strive to find the one that clears the most power.

3-Node Grid Example



Supply offers

Generator	Node	Offer	Max MW
Gen 1	A	\$5.00	2.0
Gen 2	B	\$10.00	10.0
Gen 3	C	\$7.50	0.5

Line limits

Line	Limit
$A \to C$	3 MW
$A \to B$	3 MW
$B \to C$	3 MW

Demand bids

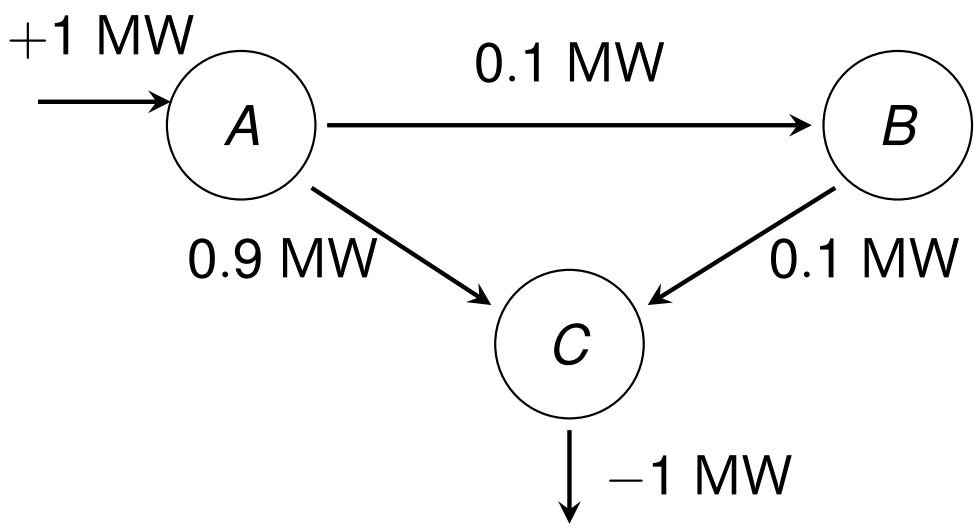
Load	Node	Demand	Bid
Load 1	A	1.0	\$6.50
Load 2	C	2.0	\$7.50
Load 3	C	1.5	\$10.00

Base Case Clearing

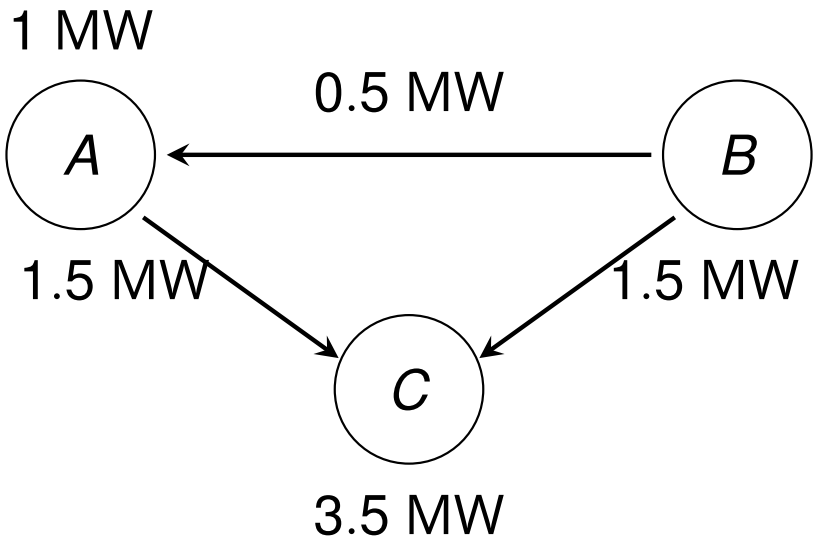
Shift factors for source A, sink C

$A \to B$	$B \to C$	$A \to C$
0.1	0.1	0.9

1 MW injected at A and withdrawn at C flows as follows:



One feasible solution

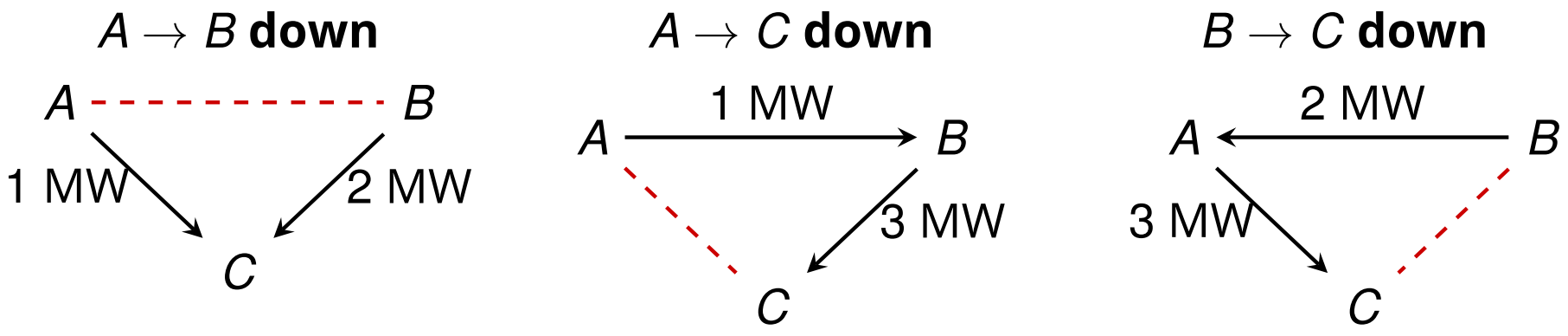


- Net loads: $A = 1$ MW, $C = 3.5$ MW.
- Gen 1 produces 2 MW.
- Gen 2 produces 2 MW.
- Gen 3 produces 0.5 MW.
- All loads are fully filled.

(N-1) Contingency Analysis

Reliability requirement

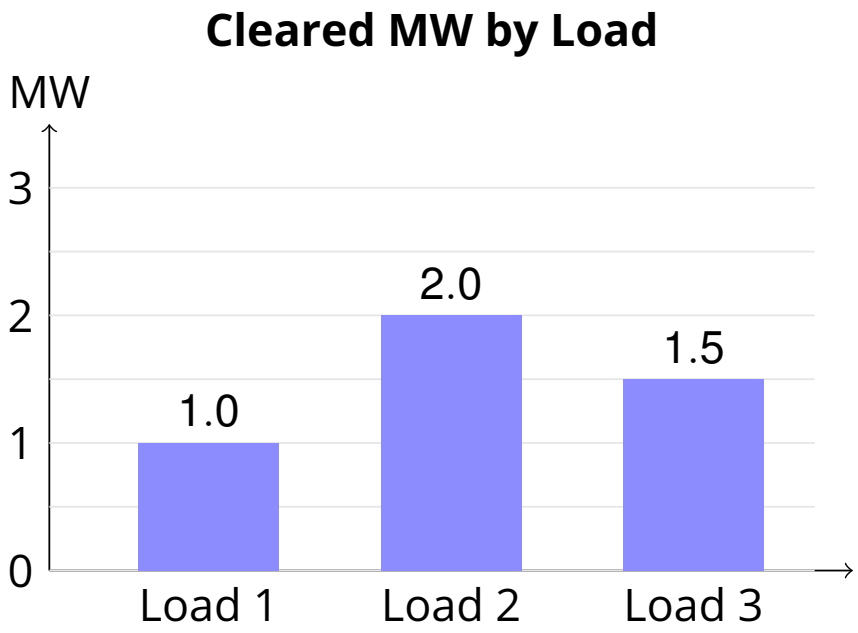
In addition to the base case, the cleared schedule must remain feasible in every contingency scenario where one transmission line goes down.

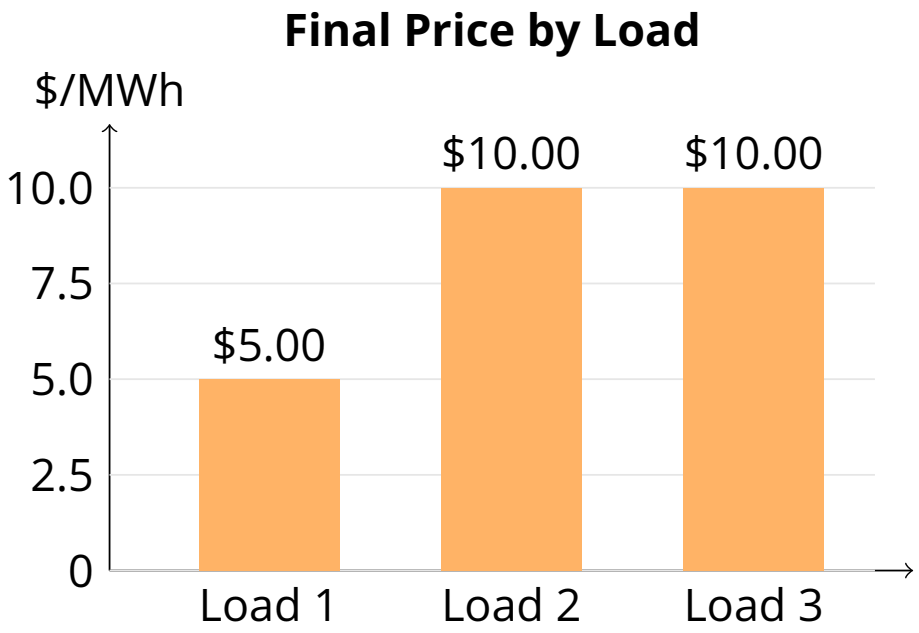


Feasible Schedule

Since this load schedule is feasible in each contingency scenario (no transmission line is over capacity), the RTO approves.

Clearing Results





Optimal dispatch

Generator	MW
Gen 1 (A)	2.0
Gen 2 (B)	2.0
Gen 3 (C)	0.5

Locational prices

Node	$/MWh
A	5.00
B	10.00
C	10.00

Takeaway

Locational prices can vary in a power grid due to transmission limits, known as congestion.

DA LP Formulation

Objective (x = cleared MWh)

max $\sum$ Load MWh $\times$ Price - $\sum$ Gen MWh $\times$ Price

s.t.

Power Balance Constraint

MW Generated = MW Consumed across the grid (assume no loss)

Gen, Load Feasibility

Generators cleared in [min, max] range. Loads allocated $\leq$ Demand

Base-Case Thermal Constraints

Cleared solution does not result in overloaded transmission line

(N-1) Contingency Thermal Constraints

No transmission overload in any contingency scenario

Financial Transmission Rights

$$\text{Payout} = (LMP_{\text{sink}} - LMP_{\text{source}}) \times \text{Quantity}$$

- An FTR is a financial right on congestion between two nodes.
- Utilities and generators use FTRs to hedge uncertain nodal price differences.
- The auction clears awarded quantities x_r subject to a Simultaneous Feasibility Test (SFT).

$$-F_l^{\max} \leq \sum_{r \in \mathcal{R}} \left(PTDF_{l,t(r)}^{(k)} - PTDF_{l,o(r)}^{(k)}\right) x_r \quad \leq F_l^{\max}$$

Takeaway 3

The RTO can only sell an FTR portfolio that remains feasible with the same transmission limits used in the Day-Ahead market.

FTR Clearing LP Formulation

Objective (x = cleared MWh)

max value of cleared FTR contracts

s.t.

Availability Constraint

Cleared quantity does not exceed number of available FTR contracts

Base-Case Thermal Constraints

Cleared solution obeys full-grid thermal constraints

(N-1) Contingency Thermal Constraints

Cleared solution obeys all contingency situations

Data Collection

Great! Let's just run this on real grid data...

FERC Order No.'s 643/833

Requests for electronic copies of grid data are subject to approval by the Federal Energy Regulatory Commission and signature of NDAs.

- Luckily - the IEEE has sample networks from the 60's available in the public domain.
- We use a software called PowerWorld to read these binaries and export .csv's of bus, branch, generator definitions.
- Powerworld also calculates shift factors with transmission line characteristics from the model.

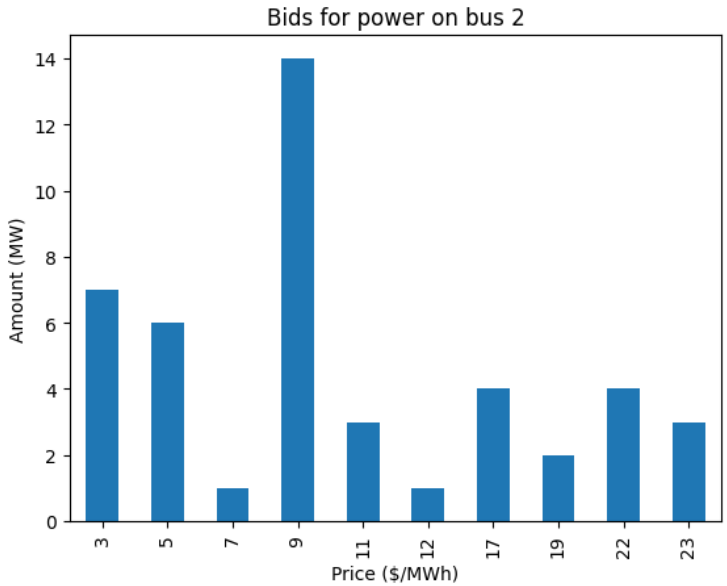
Running the LP

	From Number	From Name	To Number	To Name	Circuit	Status	Branch Device Type	Xfrmr	R	X	B	Lim MVA A	Lim MVA B	Lim MVA C
0	1	Riversde	2	Pokagon	1	Closed	Line	NO	0.0303	0.0999	0.02540	260.0	0.0	0.0
1	1	Riversde	3	HickryCk	1	Closed	Line	NO	0.0129	0.0424	0.01082	520.0	0.0	0.0

Branch Definitions from PowerWorld

Run summary

1. Chose IEEE 118-bus model for substantial size
2. Produced random bid/offer stack due to lack of data
3. Gurobi solves resulting DA LP after 19 seconds
4. FTR LP solved < 1s



Randomly generated bids for 1 bus

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Solved in 1074 iterations and 19.02 seconds (61.09 work units)
Optimal objective  7.558303852e+04
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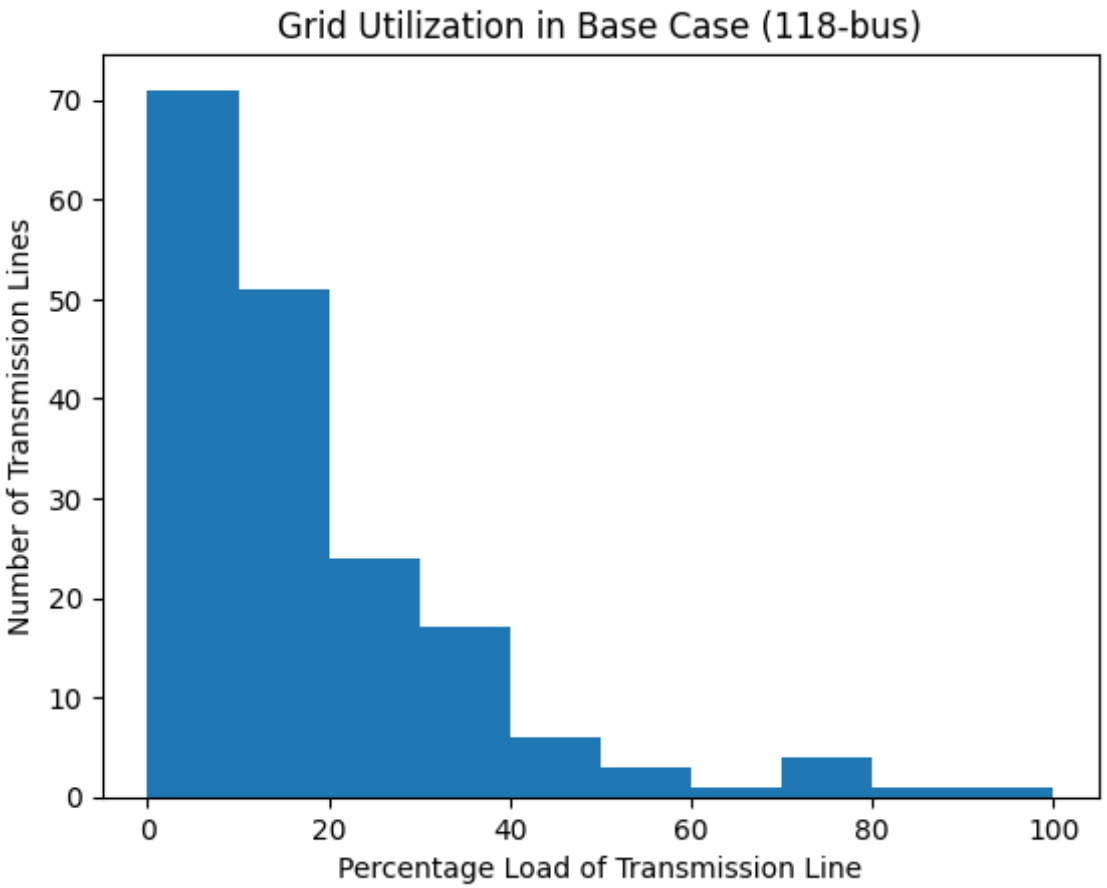
Final Gurobi Solve Time of 19s

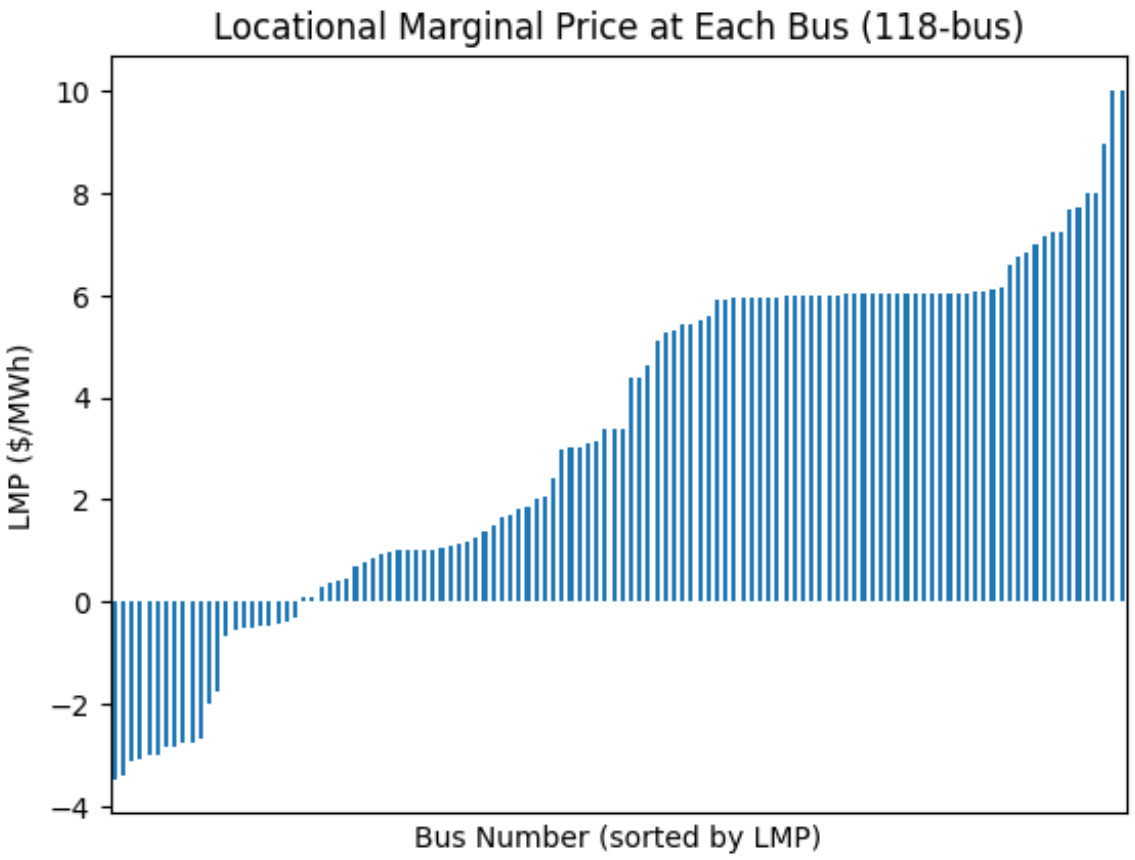
Results



Low Utilization - good for reliability

There are however some high-load lines. This is where the RTO would build new infrastructure.

Results



High LMP differences

Differences in LMP mainly come from the contingency analysis. Negative prices are also interesting!

Solving this problem at scale

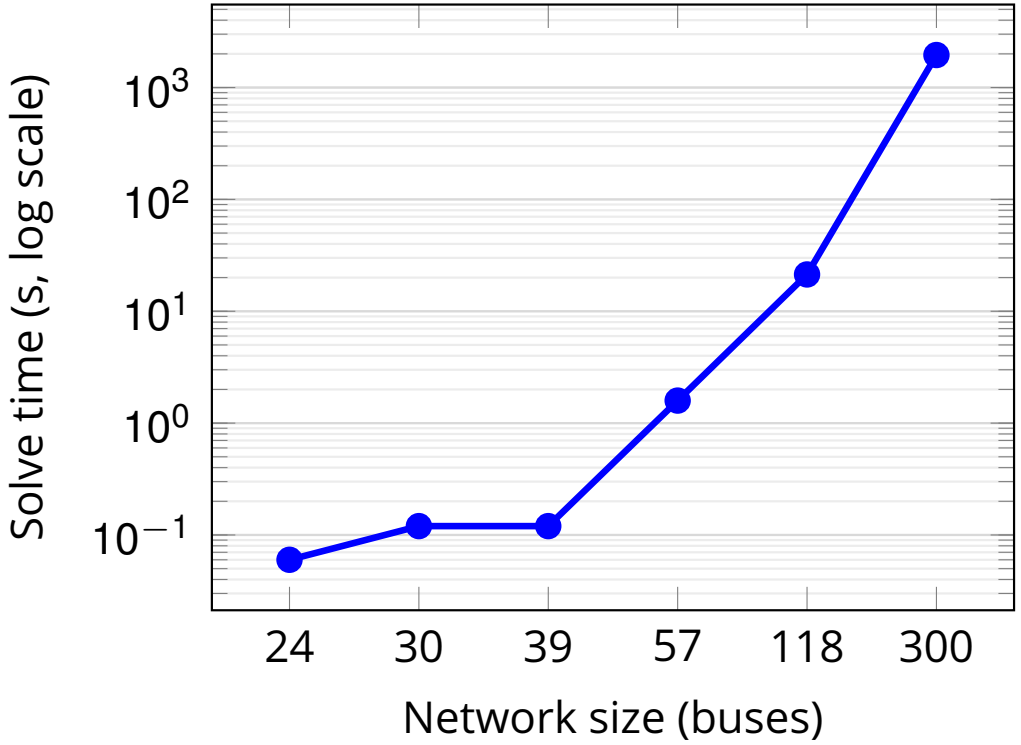
PJM in 2024

- 1,436 generators
- 6,000+ buses
- 142,158 km of transmission lines
- 800 TWh transmitted, about 53,000 Artemis II missions

PJM runs a similar LP every 5 minutes.

Source: Wikipedia: PJM Interconnection

Observed solve time on our machine (log scale)



Thank you

Power Flow v. Min Cost Flow

- Power cannot be manually routed line-by-line; it spreads according to the electrical properties of the network.
- PTDFs (shift factors) measure how a 1 MW injection at one node and withdrawal at another loads each transmission line.
- The RTO must enforce thermal limits in the base case and after any single-line outage.

$$\text{flow}_l = \sum_{s \in \mathcal{N}} PTDF_{l,s} \, \text{inj}_s$$

Reliability requirement

The same dispatch must remain feasible under all monitored contingencies (N-1 security).

DA LP Formulation

$$\max_{g,d} \quad \sum_{d \in \mathcal{D}} p_d \, d_d \; - \; \sum_{g \in \mathcal{G}} c_g \, g_g$$

$$\text{s.t.} \quad \sum_{g \in \mathcal{G}} g_g \; - \; \sum_{d \in \mathcal{D}} d_d \; = \; 0$$

$$g_g^{\min} \leq g_g \leq g_g^{\max} \qquad \forall g \in \mathcal{G}$$

$$0 \leq d_d \leq d_d^{\text{nom}} \qquad \forall d \in \mathcal{D}$$

$$inj_s \; = \; \sum_{g:n(g)=s} g_g \; - \; \sum_{d:n(d)=s} d_d \qquad \forall s \in \mathcal{N}$$

$$-F_l^{\max} \leq \sum_{s \in \mathcal{N}} PTDF_{l,s} \, inj_s \leq F_l^{\max} \qquad \forall l \in \mathcal{L}$$

$$-F_l^{\max} \leq \sum_{s \in \mathcal{N}} PTDF_{l,s}^{(k)} \, inj_s \leq F_l^{\max} \qquad \forall (l,k) \in \mathcal{C}$$

1. Where $\mathcal{G}$, $\mathcal{D}$ are generators and loads

LMP Calculation

$$\lambda := \text{dual variable of the power-balance constraint,}$$

$$\mu_l^{\max},\ \mu_l^{\min} := \text{dual variables of } \sum_{s\in\mathcal{N}} PTDF_{l,s}\, inj_s \le F_l^{\max} \text{ and } \sum_{s\in\mathcal{N}} PTDF_{l,s}\, inj_s \ge -F_l^{\max},$$

$$\mu_{l,k}^{\max},\ \mu_{l,k}^{\min} := \text{dual variables of } \sum_{s\in\mathcal{N}} PTDF_{l,s}^{(k)}\, inj_s \le F_l^{\max} \text{ and } \sum_{s\in\mathcal{N}} PTDF_{l,s}^{(k)}\, inj_s \ge -F_l^{\max}.$$

$$\mathrm{LMP}_i^{\text{base}} = \lambda + \sum_{l\in\mathcal{L}} PTDF_{l,i}\left(\mu_l^{\max} + \mu_l^{\min}\right), \qquad \forall i \in \mathcal{N},$$

$$\mathrm{LMP}_i = -\left(\mathrm{LMP}_i^{\text{base}} + \sum_{(l,k)\in\mathcal{C}} PTDF_{l,i}^{(k)}\left(\mu_{l,k}^{\max} + \mu_{l,k}^{\min}\right)\right), \qquad \forall i \in \mathcal{N}.$$

FTR LP Formulation

$$\max_{x} \quad \sum_{r \in \mathcal{R}} p_r \, x_r$$

$$\text{s.t.} \quad 0 \leq x_r \leq \bar{q}_r \qquad \forall r \in \mathcal{R}$$

$$-F_l^{\max} \leq \sum_{r \in \mathcal{R}} \left(PTDF_{l,t(r)} - PTDF_{l,o(r)} \right) x_r \leq F_l^{\max} \qquad \forall l \in \mathcal{L}$$

$$-F_l^{\max} \leq \sum_{r \in \mathcal{R}} \left(PTDF_{l,t(r)}^{(k)} - PTDF_{l,o(r)}^{(k)} \right) x_r \leq F_l^{\max} \qquad \forall l \in \mathcal{L}, \; \forall k \in \mathcal{K}$$